

NAGAVARSHINI M

9597436684 | 25mt0298@iitism.ac.in
<https://www.linkedin.com/in/nagavarshini-m>

PERSONAL STATEMENT

- Energy and Environmental Engineer with hands-on experience in Scope 1 and 2 GHG inventory, life cycle assessment, and carbon credit project development. Combines field-level ESG data collection with technical modelling in OpenLCA, QGIS and Google Earth Engine, having delivered a workflow framework for a state-wide waste management programme and quantified facility-level emissions against ESG reporting frameworks. Seeking a client-facing sustainability consulting role in GHG accounting, BRSR readiness, or resource efficiency assessment.

TECHNICAL SKILLS

- Programming Languages: Python, R, C, MATLAB, Swift
- Data & Analytics: Advanced Excel (VLOOKUP, PivotTables, Power Query), Power BI, statistical analysis, pandas, NumPy, Matplotlib
- Geospatial & Simulation: QGIS, Google Earth Engine, OpenLCA, PVsyst

KEY ACHIEVEMENTS

- Secured second position in Paper presentation under Youth category on theme "Heavy Metal Remediation through Biochar Constructed Wetland" at 56th Indian Water Works Association (IWWA) Annual Convention held at Bharathiar University, Coimbatore.
- Designed a user-friendly application called 'Energy Metrica' for energy conservation using SWIFT UI (Team size – 5) and represented my team in presenting the work to the Apple India team.
- Shortlisted for the Final Presentation Round of Tata Steel M. Tech Internship Program 2026

EDUCATION

Master of Technology in Environmental Science and Engineering (CGPA: 9.42 up to second semester)

IIT (ISM) Dhanbad, Jharkhand – 826004 (2025-2027)

B. Tech Energy and Environmental Engineering

Agricultural Engineering College and Research Institute, TNAU, Coimbatore (2020-2024)

CGPA: 8.58 (Department II Rank)

GATE 2026 ES AIR 369 & 2025 AIR 634

EXPERIENCE

Intern - Sustainability and Climate change at Beyond Sustainability, Coimbatore (May 2026 – July 2026)

- Conducted research on **voluntary compliance carbon markets**, carbon credit mechanisms, and project eligibility under international standards.
- Developed a **workflow framework for the Thooimai Mission**, supporting sustainable waste management and carbon credit development.

Intern trainee, TVS Rubber Ltd, Madurai (December 2025)

- Studied boiler operations, visited STP, ETP and conducted **performance assessment** in the Maintenance Department.
- Performed Scope 1 & Scope 2 emissions quantification and aligned facility-level metrics with ESG reporting frameworks.

Research intern SusPaani Lab, Indian Institute of Science (IISc), Bangalore (June 2024 – August 2024)

- Worked on nitrification of wastewater in a Packed Bed Biofilm Reactor and a pilot-scale Fixed Bed Reactor.
- Conducted periodic monitoring of pH, conductivity, and dissolved oxygen using a HACH HQ 2200 meter; quantified ammonium-N, nitrite-N, and nitrate-N via UV-Vis Spectrophotometer following ISO/APHA methods.

Intern, Rajapalayam Mills (2023)

- Gained an overview of Textile industry processes and operations – basic training in Environment, Health and Safety-aspects.

Content Writing intern at Dpurpose Foundation NGO (Part-time) (Jan 2023 – March 2025)

- Wrote engaging blogs on sustainable entrepreneurial innovations and success stories of young innovators from Hack4Purpose, the foundation's annual flagship event.

MAJOR PROJECTS:

'Life Cycle Assessment of Grid-connected Solar Photovoltaic Systems: A Case Study of India' (Ongoing)

'Sustainable Waste Water Treatment using Biochar Constructed Wetland'

- Designed and fabricated a laboratory-scale biochar constructed wetland for heavy-metal removal, turbidity and TDS reduction.
- Treated effluent met permissible discharge limits for irrigation, car washing, and other non-potable reuse applications.

MINOR PROJECTS:

- **Site Suitability Analysis for Utility-Scale Solar Farm Development, Dhanbad District, Jharkhand:** Built a multi-criteria decision analysis model in QGIS integrating slope, solar irradiance, land use/land cover, and proximity to roads, transmission lines and settlements; reclassified and weighted each criterion layer to produce a ranked suitability map identifying high-suitability land across the district.
- **Comparative Urban Heat Island Analysis of Four Indian Cities (Delhi, Chennai, Bangalore, Jaipur):** Extracted multi-temporal Landsat land surface temperature data (2014–2026) via Google Earth Engine across four distinct climate zones; built a MATLAB pipeline for data ingestion, per-city linear regression modelling, and automated generation of comparative trend visualisations. Identified a persistent negative UHI intensity in Chennai, indicating an Urban Cool Island driven by a hotter rural hinterland.
- **Supply Chain Management of Agro-Residues, West Bengal:** Mapped district-level agro-residue availability in QGIS and modelled collection catchments and transport distances in Excel to evaluate feasible siting for biomass processing facilities.
- Fabrication of 2'x2' Portable Solar Dryer

INDUSTRIAL TRAININGS AND WORKSHOPS

- Online training programme on “Biodegradable Waste Management in Global South: Challenges and Opportunities” organized by the Centre for Science and Environment, New Delhi from 30th July – 2nd August 2024.
- Advanced Excel training programme: Analysed Rainfall & Weather data using VLOOKUP, PivotTable, Power Query and other excel tools
- One-week intensive training on Energy Audit at South India Textile Research Association (SITRA), Coimbatore Hands-on training on energy auditing methods, instruments and estimation of pay-back period
- Two-weeks training on Solar Energy Technology by Visiting Professor of Edinburgh Napier University, UK.

CERTIFICATIONS

- Environmental Policy and Action Youth Fellowship (EPAYF Cohort 2.0), IMPRI, New Delhi (2025)
- ESG Sustainability Reporting Course organized by World Environment Council (Jan 2026)
- One-Month course on “Environmental Impact Assessment and Environmental Management Plan for Development Projects” organized by CSEE from November 1st to 30th, 2025.
- Global online certificate course on “Demystifying Environmental and Sustainability Data for Effective Communication in the 21st Century” organized by the School of Environmental Communications, Centre for Science and Environment from June 27 – July 11, 2024.
- NPTEL Online Certification on “Hydrogen Energy: Production, Storage, Transportation and Safety”

EXTRA-CURRICULARS AND CO-CURRICULARS

- Student Representative, Agricultural Engineering Alumni Association of Tamil Nadu Agricultural University (2022-2024)
- Souvenir Committee member of Ecofest'23, an International Conference on Advancements in Energy Transitions and Climate Neutrality organised by AEC&RI, Coimbatore
- Vice-Captain, Orange House, TNAU (2023)

PUBLICATIONS

- *“India’s Energy Sector Decarbonisation: Progress, Challenges and Future Pathways”*, ISBN: 978-81-967390-0-3, IMPRI Impact and Policy Research Institute, New Delhi
- *“Microbial Remediation of Heavy Metals”* Abstract No.85, National Conference on “Sustainable Management of Water and Waste Water: Latest Developments” 56th IWWA Annual Convention
- *“IoT Technologies for Environmental Pollution Prevention and Improvement” Poster No. ES-IT-POS-4*, International Conference on Emerging Green Energy Technologies and Environmental Sustainability, Ecofest 2021.

PROFESSIONAL STUDENT MEMBERSHIPS AND VOLUNTEERSHIPS

Indian Water Works Association (IWWA) | Indian Society of Agricultural Engineers (ISAE)
Active NSS Volunteer (2020-2023) | SSL Student Teaching Volunteer (2022) | Volunteer
at Biodiversity and Environmental Sustainability (BEST) NGO (2024-2025)

LANGUAGES: English, Tamil, Hindi, Telugu, French (beginner)